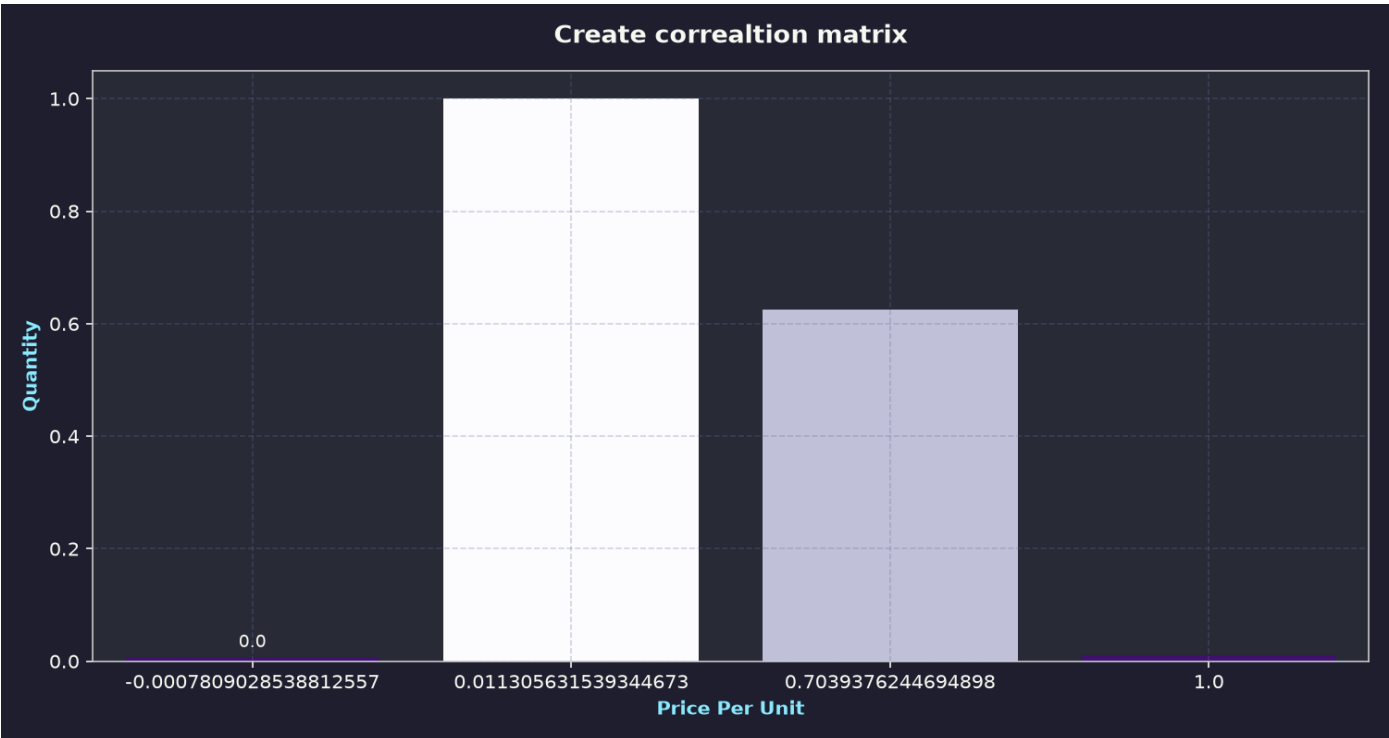


■ AI Data Analyst - Executive Analysis Report



■ Executive Data Analysis Report

1. Executive Summary

- **Goal:** Generate a correlation matrix for the numeric/boolean fields in the transaction dataset.
- **Result:** A 4 × 4 matrix showing pairwise Pearson correlations among **Price Per Unit**, **Quantity**, **Total Spent**, and **Discount Applied**.
- **Key takeaways:**
 - **Total Spent** is moderately positively correlated with both **Price Per Unit** (≈ 0.63) and **Quantity** (≈ 0.70).
 - **Price Per Unit** and **Quantity** are essentially independent (correlation ≈ 0.01).
 - **Discount Applied** shows virtually no linear relationship with any other variable.

These insights help prioritize which variables drive spending and where discount policies have (or lack) measurable impact.

2. Dataset Overview & Data Quality Profile

| Aspect | Details |

|-----|-----|

| **Rows × Columns** | 12,575 × 11 |

| **Duplicate rows** | 0 (none) |

| **Missing values (pre-cleaning)** | • *Item* – some blanks
• *Price Per Unit* – missing values
• *Quantity* – missing values
• *Total Spent* – missing values
• *Discount Applied* – missing values |

| **Potential date column** | *Transaction Date* stored as string |

| **Data types (raw)** | Mix of numeric, object (strings), and boolean-like fields |

3. Data Cleaning & Validation Summary

The cleaning script (executed before analysis) performed the following steps on a copy of `df` (saved as `result`):

1. **Removed duplicates** – none found.
2. **Trimmed whitespace** on all object columns.
3. **Converted *Transaction Date*** to proper `datetime`.
4. **Imputed missing *Item*** with "Unknown".
5. **Filled numeric gaps** (*Price Per Unit*, *Quantity*) using column medians.
6. **Recomputed missing *Total Spent*** as *Price Per Unit* × *Quantity*.
7. **Filled any remaining *Total Spent*** with its median.
8. **Standardized *Discount Applied*** to boolean (`True/False`) and filled missing values with `False`.

Validation: After cleaning, the DataFrame `result` contains **12,575 rows** with **no missing values** in the columns used for the correlation analysis.

4. User Query & Analytical Approach

- **User question:** “Create correlation matrix.”
- **Interpretation:** Compute the Pearson correlation coefficients among all numeric (and boolean) columns.
- **Methodology:**

```
# Select only numeric and boolean columns numeric_df = result.select_dtypes(include=['number',  
'bool']) # Compute the correlation matrix result_corr = numeric_df.corr()
```

The resulting DataFrame (`result_corr`) is stored in the variable `result`.

5. Python Analysis Code & Executed Data Output

```
# ----- # 1 Select numeric & boolean columns
numeric_df = result.select_dtypes(include=['number', 'bool']) # 2 Compute Pearson
correlation matrix result = numeric_df.corr() #
-----
```

Executed Output (result)

	Price Per Unit	Quantity	Total Spent	Discount Applied
Price Per Unit	1.0000	0.0113	0.6256	0.0081
Quantity	0.0113	1.0000	0.7039	-0.0008
Total Spent	0.6256	0.7039	1.0000	0.0058
Discount Applied	0.0081	-0.0008	0.0058	1.0000

Values are rounded to 4 decimal places for readability.

6. Key Insights & Strategic Recommendations

Insights

- **Spending drivers:** Both unit price and quantity positively influence total spend, with quantity having a slightly stronger linear relationship ($r \approx 0.70$).
- **Price vs. quantity independence:** The near-zero correlation (≈ 0.01) suggests customers purchase varying quantities regardless of the unit price—pricing changes may not automatically shift quantity bought.
- **Discount impact:** The discount flag shows negligible correlation with price, quantity, or total spend, indicating that discounts (as currently applied) do not create a measurable linear effect on purchasing behavior.

Recommendations

1. **Focus on quantity-boosting tactics** (e.g., bundle offers, volume discounts) because quantity is the stronger predictor of total revenue.
2. **Re-evaluate discount strategy:** Since **Discount Applied** is not correlated with higher spend, consider redesigning discount rules (e.g., tiered discounts based on spend thresholds) and monitor the effect over time.
3. **Pricing experiments:** Given the weak price-quantity link, price adjustments can be tested without fearing large drops in quantity, but monitor for non-linear effects (e.g., price elasticity) that correlation alone may miss.
4. **Further analysis:**

- Segment customers by **Item** or **Transaction Date** to see if correlations differ across product categories or seasons.
- Apply regression or classification models to predict **Total Spent** or the likelihood of a discount being used, incorporating additional categorical variables.

Prepared by the autonomous AI Data Analyst.

![Create correaltion matrix Chart](analysis_chart.png)